

Project: Modeling DNA Sequence Determinants of Regulatory Activity

Introduction:

The precise control of gene expression is fundamental to biological systems. DNA sequences harbor regulatory elements that govern the transcriptional output of genes. Understanding the sequence-function relationship of these elements is critical for elucidating complex biological processes and the molecular basis of disease.

This project challenges you to investigate the predictive power of machine learning in modeling DNA sequence determinants of regulatory activity. Drawing inspiration from established computational approaches such as **DeepSTARR** and **DeepMEL**, you will develop and evaluate regression and classification models to predict quantitative and qualitative measures of regulatory potential.

The Task:

You will be provided with a dataset comprising DNA sequences and associated regulatory information:

- **is_active**: A binary label (0 or 1) indicating the presence or absence of regulatory activity for a given sequence. You are required to develop a binary classification model for this endpoint.
- **rna_dna_ratio**: A continuous variable representing the experimentally determined ratio of RNA to DNA levels associated with the sequence, serving as a quantitative measure of regulatory strength. You are required to develop a regression model to predict this value.

You may explore shared or independent model architectures for these two prediction tasks, where your input is the **sequence** column.

Evaluation:

Your work will be evaluated during the lab sessions on **April 24th and 25th**, but you need to send me your project **24 hours before**. The evaluation will consist of two components:

1. **Performance on Hidden Test Datasets (5 points total - split)**: The performance of your submitted models will be assessed on independent, held-out test datasets for both the classification and regression tasks. Evaluation will be conducted via a command-line executable:

```
python evaluation_script.py <path_to_model> <path_to_test_data>
```

- **<path_to_model>**: The file path to your serialized trained model (e.g., .pth, .h5, .pkl).
- **<path_to_test_data>**: The file path to a .tsv (or similar delimited) file containing the test DNA sequences. This file will include columns for a unique sequence identifier (**id**) and the raw DNA sequence (**sequence**).

The **evaluation_script.py** must:

- Parse the command-line arguments to load the model and test data.
- Read the test data file, extracting the sequence information.
- Load the trained model and generate predictions for both **is_active** (outputting binary labels: 0 or 1) and **rna_dna_ratio** (outputting continuous numerical values).
- Print the predictions to standard output in a **tab-separated format** with the columns: **id**, **predicted_is_active**, **predicted_rna_dna_ratio**, maintaining the order of sequences from the input test data file.

The top 5 performing models overall, based on established metrics for both classification (e.g. F1-score) and regression (e.g. R^2), will be awarded points as follows: 5, 4, 3, 2, and 1 point for the top to fifth-best model, respectively.

2. **Presentation (5 points)**: A concise **5-minute presentation** of your methodology and findings is required. Scoring will be based on:

- **Structure (1 point)**: Logical organization and clarity of the presentation.
- **Methodological Rigor (0-3 points)**: Soundness and appropriateness of the analytical methods employed for model development, validation (e.g., cross-validation), and comparison.
- **Interpretation and Discussion (1 point)**: Insightful discussion of the results in the context of limitations of the developed models, and potential avenues for future research.

Deliverables:

You are required to submit:

- All code developed for model training and evaluation.
- The serialized trained model file.
- The **evaluation_script.py** adhering to the specified command-line interface and output format.
- A brief report summarizing your approach, key results, and discussion points for your presentation.

Emphasis on Evaluation Script: Ensure the **evaluation_script.py** is robust, well-documented, and accurately implements the loading of your model and the processing of the test sequences to produce the required tab-separated output.

This project provides an opportunity to apply machine learning principles to a significant problem in computational biology. I am looking forward to seeing your creative solutions!